

## 5. Contrast of Formulations

From the preceding discussion, it was seen that the  $Ri$  formulation is equivalent to a  $\zeta$  formulation. However, in finite-difference models, the lowest level is usually treated differently. Equations (1a,b), (2), (4), and (6) can be combined to produce the following relations

$$K_m \frac{\partial u}{\partial z} = u_*^2 \cos \alpha \quad (44a)$$

$$K_m \frac{\partial v}{\partial z} = u_*^2 \sin \alpha \quad (44b)$$

$$K_h \frac{\partial \theta}{\partial z} = u_* \theta_* \quad (44c)$$

where  $\tan \alpha = (\partial v / \partial z) / (\partial u / \partial z)$ . Expressed separately for momentum ( $z_{0m}$ ) and heat/scalars ( $z_{0h}$ ), the friction parameters in the lowest level are

$$u_* = \frac{\kappa |\mathbf{V}_1|}{\ln \frac{z_1}{z_{0m}} - \psi_m(\zeta_1)}, \quad \psi_m(\zeta) = \int_0^\zeta \frac{1 - \phi_m(\xi)}{\xi} d\xi \quad (45a)$$

$$\theta_* = \frac{\kappa(\theta_1 - \theta_g)}{\ln \frac{z_1}{z_{0h}} - \psi_h(\zeta_1)} = \frac{\kappa(\theta_1 - \theta_g)}{\ln \frac{z_1}{z_{0m}} + kB^{-1} - \psi_h(\zeta_1)}, \quad \psi_h(\zeta) = \int_0^\zeta \frac{1 - \phi_h(\xi)}{\xi} d\xi, \quad (45b)$$

where  $z_1$  is the height of the lowest model level,  $\theta_g$  is the surface skin temperature, and  $kB^{-1} \equiv \ln(z_{0m}/z_{0h})$  parameterizes the excess thermal resistance at the surface (see Panofsky, 1963; Garratt, 1992).

Note that the lower limit of integration of (45a,b) should technically be  $z_{0m}$  and  $z_{0h}$ , but the approximation  $z_0 = 0$  is usually made. The mathematical evaluation of the integrals in (45a,b) is straightforward in the SBL because the shear functions are linear. The process is more difficult in the UBL (see Paulson, 1970; Nickerson and Smiley, 1975; Benoit, 1977). Modern surface layer schemes recognize that assuming equal roughness lengths ( $z_{0m} = z_{0h}$ ) is physically inaccurate due to the absence of pressure drag for scalar transport at the microscale (Blyth and McIlveen, 1993).

Calculating the friction parameters from (45a,b) requires an iterative procedure because the upper limit of integration  $\zeta$  also depends upon the friction parameters. This time-consuming procedure can be circumvented in a Richardson number formulation. However, a forward difference approximation to the lowest-layer  $Ri$  produces the bulk Richardson number,

$$\text{Ri}_b = Bz_1 \frac{\theta_1 - \theta_g}{|\mathbf{V}_1|^2} \quad (46)$$

which does not depend directly upon  $z_{0m}$  or  $z_{0h}$  (although it should), but does depend upon the ambiguous surface skin temperature. Recall that the  $\zeta$  formulation incorporated roughness length dependence by means of logarithmic transformations and the chain rule:

$$\frac{\partial}{\partial z} = \frac{d \ln(z/z_{0m})}{dz} \frac{\partial}{\partial \ln(z/z_{0m})} = \frac{1}{z} \frac{\partial}{\partial \ln(z/z_{0m})} \quad (47)$$

Thus, equations (16a) and (16b) in centered finite-difference form become

$$u_* = \sqrt{f_m(\text{Ri}_{1/2})} \frac{\kappa |\mathbf{V}_1|}{\ln\left(\frac{z_1}{z_{0m}}\right)} \quad (48a)$$

$$\theta_* = \frac{f_h(\text{Ri}_{1/2})}{\sqrt{f_m(\text{Ri}_{1/2})}} \frac{\kappa(\theta_1 - \theta_g)}{\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1}} \quad (48b)$$

Logically,  $\text{Ri}$  must also undergo a dual logarithmic transformation for momentum and scalar profiles:

$$\text{Ri} = B \frac{\partial \theta}{\partial z} \bigg/ \left| \frac{\partial \mathbf{V}}{\partial z} \right|^2 = B \frac{1}{z} \frac{\partial \theta}{\partial \ln(z/z_{0h})} \bigg/ \left[ \frac{1}{z^2} \left| \frac{\partial \mathbf{V}}{\partial \ln(z/z_{0m})} \right|^2 \right] \quad (49)$$

Evaluating (49) across the surface layer yields the revised lower-layer Richardson number:

$$\text{Ri}_{1/2} = Bh_1 \frac{\theta_1 - \theta_g}{|\mathbf{V}_1|^2} \quad (50)$$

where the effective layer scale  $h_1$  explicitly incorporates thermal roughness via  $kB^{-1}$ :

$$h_1 = \bar{z}_m e^{\frac{1}{2}kB^{-1}} \frac{\left[ \ln\left(\frac{z_1}{z_{0m}}\right) \right]^2}{\ln\left(\frac{z_1}{z_{0m}}\right) + kB^{-1}} \quad (50a)$$

with geometric mean height  $\bar{z}_m = \sqrt{z_1 z_{0m}}$ . When momentum and thermal roughness lengths are assumed equal ( $z_{0m} = z_{0h} \implies kB^{-1} = 0$ ),  $h_1$  simplifies to  $\sqrt{z_1 z_{0m}} \ln(z_1/z_{0m})$ .

Finally, the closed-form  $\text{Ri}$  formulation is fully equivalent to the  $\zeta$  formulation while retaining explicit roughness parameterizations. Combining (48a,b) yields the surface fluxes:

$$u_*^2 = \frac{\kappa^2 f_m(\text{Ri}_{1/2}) |\mathbf{V}_1|^2}{\left[ \ln \left( \frac{z_1}{z_{0m}} \right) \right]^2} \quad (51a)$$

$$u_* \theta_* = \frac{\kappa^2 f_h(\text{Ri}_{1/2}) |\mathbf{V}_1| (\theta_1 - \theta_g)}{\ln \left( \frac{z_1}{z_{0m}} \right) \left[ \ln \left( \frac{z_1}{z_{0m}} \right) + kB^{-1} \right]} \quad (51b)$$

which can be used in conjunction with (44a,b,c) to evaluate momentum and sensible heat fluxes without numerical iteration.

### Summary of Refinements Made to Section 5

- **Roughness Length Separation:** Replaced generic  $z_0$  with distinct momentum roughness ( $z_{0m}$ ) and thermal/scalar roughness ( $z_{0h}$ ).
- **Excess Thermal Resistance Integration:** Introduced  $kB^{-1} \equiv \ln(z_{0m}/z_{0h})$  into equations (45b), (48b), and (51b), bringing the formulation into alignment with modern land-surface models (e.g., Noah-MP, CLM).
- **Updated Effective Layer Height Scale ( $h_1$ ):** Generalized equation (50) into (50a) so that  $\text{Ri}_{1/2}$  accounts for the differential vertical gradients caused by  $z_{0m} \neq z_{0h}$ .